WW-T-700/1E
29 December 1983
SUPERSEDING
WW-T-700/1D
March 13, 1967

FEDERAL SPECIFICATION SHEET

TUBE, ALUMINUM, DRAWN, SEAMLESS, 1100

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring seamless tube drawn from aluminum 1100 described herein shall consist of this document and the latest issue of WW-T-700/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for seamless tube drawn from aluminum 1100.

1.2 Classification.

1.2.1 Tempers. The drawn seamless tube shall be of the following tempers: O, H12, H14, H16, H18, and F, as specified (see 6.2 and 6.3). The definitions of these tempers shall be as specified in WW-T-700/GEN.

1.2.2 Types. The tube shall be of the following types:

Type		Appearance
I	-	Round
II	-	Rectangular and square
III	-	Streamline
IV	-	Oval
V	-	Odd shapes

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

WW-T-700/GEN - Tube, Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification for

FSC 4710

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Silicon plus iron	-	0.95
Copper	0.05	0.20
Manganese	-	0.05
Zinc	-	0.10
Other elements, each	-	0.05
Other elements, total 2/	-	0.15
Aluminum	99.00	-

1/ Except for "Aluminum" and "Others", analysis normally is made for elements for which specific limits are shown

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum

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3.2 Mechanical properties.

3.2.1 Tensile strength. The tensile strength parallel to the direction of drawing shall conform to the requirements specified in table II.

TABLE II. Tensile strength

Temper	Wall thickness, inch	Tensile strength, minimum, ksi
O	0.014 to 0.500, incl.	15.5 1/
H12	0.014 to 0.500, incl.	14.0
H14	0.014 to 0.500, incl.	16.0
H16	0.014 to 0.500, incl.	19.0
H18	0.014 to 0.500, incl	22.0
F	All	2/

1/ Maximum
2/ No requirements

3.2.2 Flattening. When specified (see 6.2), round tube (type 1) in O, H12, H14 and H16 tempers shall withstand, without cracking, the flattening test or the alternative bend test specified in WW-T-700/GEN. The values for flattening factor "F" are specified in table III.

TABLE III. Flattening factor

Temper	Wall thickness, inch	F
O	0.014 to 0.500, incl.	2
H12	0.014 to 0.500, incl.	3
H14	0.014 to 0.500, incl.	6
H16	0.014 to 0.500, incl.	8

3.2.2.1 Alternative bending factor "N". The values for the alternative bending factor "N" are specified in table IV.

TABLE IV. Bending factor

Temper	N
H12	1
H14	4
H16	6

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3.2.3 Flaring. When specified (see 6.2), round tube (type I) in the H14 temper shall be capable of being flared as specified in WW-T-700/GEN.

4. QUALITY ASSURANCE PROVISIONS (see WW-T-700/GEN)

5. PREPARATION FOR DELIVERY (see WW-T-700/GEN)

6. NOTES

6.1 Intended use. This tube is intended for use when a very good formability is required, and when high strength is not necessary. This tubing is not applicable for use in transferring hydraulic/pneumatic mediums.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Temper of material required (see 1.2.1)
(c) Type of tube required (see 1.2.2)
(d) Whether round tube (type 1) in O, H12, H14 and H16 tempers should withstand flattening (see 3.2.2)
(e) Whether round tube (type 1) in H14 temper should withstand flaring (see 3.2.3)
(f) Dimensions required
(g) Requirements for sizes not specifically covered (see WW-T-700/GEN)
(h) Selection of applicable levels of preservation and packing (see WW-T-700/GEN)

6.3 Availability of the types. Types II through V are only available in the O and F tempers. Type I is available in all tempers.

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Army-AR
Navy-OS, SH
Air Force-99
DLA-CS

User Activities

Army-MI
Navy-MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
COMMERCE-NBS
HEW-FEC
NASA-KSC, MSF
USDA-AFS

PREPARING ACTIVITY:

NAVY-AS

DoD Project 4710-0707

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